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Physics-Informed Neural Network to Estimate Downhole Torsional Vibrations and Friction Coefficients

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Abstract

This paper introduces a method to estimate bit rotation speed and friction parameters in torsional drilling systems using Physics-Informed Neural Networks (PINNs). The proposed approach is versatile and applicable in the general setting of parameter estimation for nonlinear differential equations with partial measurements. The method is validated on both simulated and field data, making, to the best of our knowledge, the first application of PINNs for parameter estimation of a dynamical system using industrial data. The approach successfully predicts downhole bit rotation speed ω_{Bit} , and identifies friction coefficients along the drill string based solely on 1Hz surface measurements. Furthermore, it is generalizable, allowing it to be utilized for any well without restrictions, thereby addressing the limitations inherent in data-based models. Unlike physical models that require friction coefficients as input to estimate ω_{Bit} , this approach relies solely on surface measurements to predict both ω_{Bit} and friction coefficients. A transfer learning technique is incorporated to expedite the training process when transitioning between different sequences of surface data. This strategy proves highly effective, both in accurately estimating the friction coefficients and in reducing training time by nearly sixfold, thereby enabling the PINN's practical application in real-time scenarios. The calibrated friction coefficients and predicted bit rotation speed, obtained over the full drilling duration, provide valuable insights for drilling engineers, enabling enhanced control and optimization of critical parameters like weight-on-bit and rate-of-penetration, thereby improving drilling performance and efficiency.

Introduction

This paper focuses on the estimation of bit rotation speed and friction parameters during drilling operations. Monitoring variations in bit rotation speed is crucial for optimizing the rate of penetration (ROP) and minimizing energy loss, thereby maximizing drilling efficiency and reducing operational time and cost. Downhole measurements, including bit rotation speed, are not always available as BHAs are not always equipped with sensors. When they do exist, they are often not available in real time and only collected after tripping operations. Alternatively, they can be transmitted to the surface in real time using pressure pulses

in the drilling fluid, though this method has low data transmission rate and accuracy, limiting its usefulness. Another option is using a wired connection integrated into the drill pipes, which provides high-speed real-time data transmission but is significantly more expensive due to the embedded transmission system.

Assessing the value of friction coefficients is key at various stages of the process. As the drill string is tripped in or out, or rotated on- or off-bottom, the friction forces must be considered. Similarly, qualitative assessments and quantitative estimates of the torque, drag, vibration intensity, and drill-ahead prediction are critically important in many phases of the well construction. During the design phase, these calculations are affected by the friction coefficients, which play an important role on the wellbore quality and in the drill string mechanics calculations (Maidla, et al., 1987; Locke, et al., 1988; Samuel, 2010). Underestimating friction coefficients may lead to a significant underestimation of frictional drag resulting in a poor definition of the operation limits or to undrillable wellbore trajectories.

Furthermore, the variation of the friction coefficients along the wellbore can generate torsional drill string vibrations (Halsey, et al., 1986; Kyllingstad, et al., 1988). The dependency of the friction forces on the slip velocity, or rotation speed in the case of torsional dynamics, is known to be a source of stick-slip vibrations. These vibrations consist of an alternation between a stick phase where the drill bit rotation is stopped and a slip phase where the bit rotation speed can reach up to two or three times the surface rotation speed (Shen, et al., 2017). Due to the cyclic nature of stick-slip, different problems can arise such as drill pipe fatigue problems (Kriesels, et al., 1999), drill string components failures, and bit damage (Mensa-Wilmot, et al., 2000).

The estimation of friction coefficients is highly complex due to their dependence on multiple factors such as temperature, drill string materials, rock formation, and drilling fluid. This complexity renders them one of the most uncertain factors in most mechanical computations concerning the drill string (Elgibaly, et al., 2017). Many studies use surface measurements combined with a simple model of torsional stick-slip oscillations to predict bit rotation speed and friction coefficients, such models for analysis and control, either distributed (Aarsnes, et al., 2018; Aarsnes, et al., 2019; Saldivar, et al., 2011) or finite-dimensional (Canudas-de-Wit, et al., 2008; Richard, et al., 2004). For friction coefficients estimation, a torque and drag model based on simple equations for friction in deviated wellbores is derived in (Johancsik, et al., 1984; Menand, et al., 2006; Dao, et al., 2019). However, the model must be calibrated before being used for each new well. Control configuration methods are developed in (Lu, et al., 2017) in order to track a desired speed and to obtain the equivalent friction torques. Nonetheless, the evaluations are focused only on simulated data. A soft sensor to estimate the friction coefficients and angular velocity using only surface measurements while drilling off-bottom is presented in (Aarsnes, et al., 2019). The method is extended for on-bottom drilling in (Auriol, et al., 2020). Based on a distributed torsional model, it relies on linear observer design (Di Meglio, et al., 2020) to construct estimates of the unmeasured states and parameters. This yields a method implementable in real time, but that fails to converge in some cases, when linear correction terms and parameter update laws are insufficient.

Machine learning models for approximating nonlinear functions have been proposed too. Such a data-driven approach predicting the friction coefficients using the measured top-drive torque and angular velocity as input is developed in (Auriol, et al., 2020), where a neural network regressor is trained using a thousand of simulated training points with different ranges of depths and drill string dimensions. However, the trained model does not generalize well on data outside of the training set, typically corresponding to new wells. To estimate the bit behavior, spectrogram analyses of surface and downhole measurements is proposed by Bernasconi and Vassallo (Bernasconi, et al., 2001). Bayesian inference combined with deep neural networks for state estimation is proposed in (Mestav, et al., 2018). However, purely data-driven models need very large training datasets, with poor physical coherence of the predicted parameters, and are unable to generalize beyond the training datasets.

To overcome these drawbacks, Physics-Informed Neural Networks (PINNs), presented in (Raissi, et al., 2019), are used to leverage both measured data and physics laws while solving a regression task. This

approach has been tested in many engineering sectors such as gas flow dynamics (Nath, et al., 2023), nano-optics and metamaterials (Chen, et al., 2020), supersonic flows (Jagtap, et al., 2022), and material modeling (Pun, et al., 2019).

This paper presents the first attempt to use physics-informed neural networks for predicting physical parameters during drilling operations. It is used to predict the rotation velocity of the drill bit and friction coefficients using only surface measurements (surface torque and surface rotation speed) at 1 Hz, which is the industry standard. When stick-slip oscillations are present, both the static and dynamic friction coefficients can be estimated. The modeling approach of this work consists of the following steps: (i) A neural network is built taking time as the input and having the vector of estimated states as the output. The network is trained using a loss function that comprises two terms: one forcing estimated states to satisfy the differential equation representing the dynamics, and one forcing the model output to match measured data. The unknown parameters appear in the differential equation and are optimized along with the weights of the neural network. This combination of supervised and unsupervised learning is referred to as Physics-Informed Neural Networks (PINN) (Cuomo, et al., 2022; Lawal, et al., 2022). (ii) This procedure is applied to the bit velocity and friction estimation problem by using a two-degrees of freedom model of torsional dynamics with Coulomb stiction. The merits of the present approach are illustrated with a large number of simulations with varying friction coefficients, and also by applying it to field data. In the absence of ground truth data for the friction coefficients, the added value of the estimated parameters is showcased by running higher-fidelity simulations incorporating them as inputs and comparing the results to recorded downhole data. (iv) Finally, a transfer learning procedure is proposed to significantly reduce the computational burden of the training process at the cost of a minor decrease in performance. The method is general, i.e. applicable to any well, thus addressing the limitations of data-based models. It is also robust, designed to improve drilling performance and mitigate stick-slip. The model is tested on real drilling data, demonstrating effectiveness even in the presence of nonlinear terms, and successfully converges from any initialization.

This paper is structured as follows: in the first section, the drilling process is briefly presented and a torsional dynamics model of the drill string is derived. In the second section, the parameter estimation strategy is shown in detail. This approach is applied on simulated and field data in the third section and the modeling results are analyzed and discussed. In the fourth section, a case study of two test wells is presented. Finally, concluding remarks are drawn in the last section.

Drilling and Friction Parameters

While drilling, a variety of downhole vibrations (axial, lateral, torsional) can appear independently or simultaneously in coupled modes. In the present paper, we focus on torsional vibrations and more specifically on the so-called stick-slip, which is a periodic low-frequency torsional oscillation, where the bit rotation speed fluctuates between zero in the stick phase, and several times the surface rotation speed in the slip phase. Stick-slip is generally caused either by the contact between the drill string and the borehole (friction-induced stick-slip), or the interaction between the drill bit and the rock formation (cutting-action-induced stick-slip). Regardless of their root cause, the alternation between stick and slip phases provides information on the friction forces acting on the system.

Drill String Torsional Dynamics Model

In this section, torsional dynamics models derived from (Dao, et al., 2019) is presented. In view of estimating friction parameters and performing simulations, the model can be approximated as a series of n discrete masses, as depicted in Fig. 1 where we assume that the drill string is rotated from the surface by a motor with rotary speed ω_0 and inertia I_0 . At the bottom extremity, torque created by the rock cutting process (torque on bit or TOB) is applied to the drill string and is considered as a function of the bit rotation speed ω_{Bit} .

Along with the well trajectory, the drill string may be in contact with the borehole. The contact forces create friction torques $\vec{T}^f$ applied to the drill string. This torque varies over time according to a friction law as a function of the drill string rotary speed at the contact point. In this lumped model, each mass behaves as a torsional pendulum with a linear spring of torsional stiffness K_i , a torsional damping C_i and a mass inertia I_i , which are connected to the top drive I_0 (Navarro-Lopez, et al., 2004). By defining $\vec{\phi} = (\phi_0, \phi_1, \dots, \phi_n)^T$ the angular displacement vector for the n masses from the surface to the bit, the equation of motion of the drill string torsional behavior can be expressed as the following matrix equation:

$$M\ddot{\vec{\phi}} + C\dot{\vec{\phi}} + K\vec{\phi} = -\vec{T}^f - \vec{T}^b \quad (1)$$

where:

- $M \in \mathbb{R}^{(n+1) \times (n+1)}$ is the diagonal inertia matrix with entries I_j , $j = 0, \dots, n$.
- $C \in \mathbb{R}^{(n+1) \times (n+1)}$ is the diagonal damping matrix with entries C_j , $j = 0, \dots, n$. We note that the drill string conventionally behaves elastically without internal damping. In this model, the damping terms come from external interactions such as viscous mud-structure friction (Dao, et al., 2019).
- $K \in \mathbb{R}^{(n+1) \times (n+1)}$ is block-diagonal stiffness matrix, whose block entries are the n sub-matrices K_i $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ describing the stiffness of each element with $K_i = \frac{G J_i}{L_i}$ where G is shear modulus, L_i is the length of the i -th mass, and J_i is its polar moment of inertia.
- T_i^f for $i \in [1, \dots, n]$ represents the Coulomb friction for the i -th mass.
- $\vec{T}^b$ represents the TOB due to the interaction between the rock and the bit. Therefore:

$$\vec{T}^b = (0, \dots, 0, T_n^b)^T \quad (2)$$

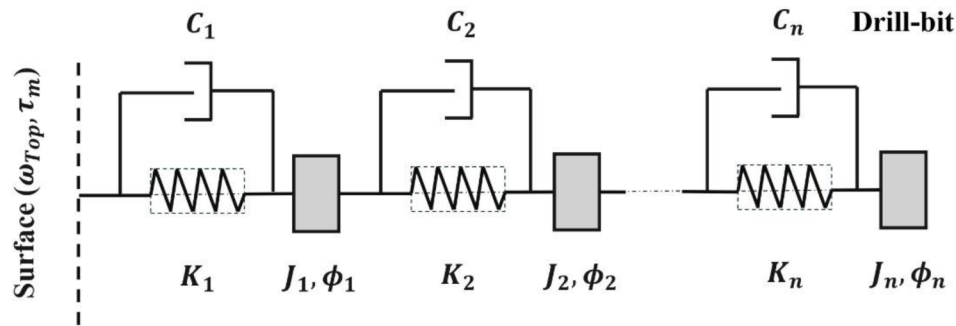


Figure 1—Drill string mass-spring system.

Torque-On-Bit and Friction Laws

Several non-linear functions have been proposed to describe the variation of the TOB and friction laws as a function of the drill bit rotation speed ω_{Bit} and the drill string contact point rotary speed ω_i , respectively (Abbassian, et al., 1998; Challamel, et al., 2000). Most of them are characterized by the presence of the so-called "Stribeck effect", which consists in a difference between the static and the dynamic values (Brett, 1992). Following (Dao, et al., 2019), both the bit-rock interaction and friction laws are modeled using a regularized Stribeck effect. More precisely, we have:

$$T_n^b = \Gamma(\omega_{\text{Bit}}, \tau_s, \tau_d) \quad T_i^f = R F_i^N \Gamma(\omega_i, \mu_s, \mu_d)$$

Where τ_s is the static torque, τ_d is the dynamic torque, R_0 is the drill string outer radius at the contact point, F_i^N is the normal contact force computed using a stiff-string Torque & Drag model (see e.g (Aarsnes, et al., 2019; Dao, et al., 2019; Belaid, 2005; Menand, et al., 2006)), μ_s is the static friction coefficient, μ_d is the dynamic friction coefficient, and the function Γ takes the following form:

$$\Gamma(\omega, \alpha, \beta) = \begin{cases} \text{sign}(\omega) \left[\beta + (\alpha - \beta) e^{\frac{-\omega}{\omega_c}} \right] & \text{if } |\omega| \geq \omega_{\text{limit}} \\ \frac{\omega}{\omega_{\text{limit}}} \Gamma(\omega_{\text{limit}}, \alpha, \beta) & \text{else} \end{cases} \quad (3)$$

In this expression, ω_c is a characteristic rotary speed, ω_{limit} is the limit rotary speed used to avoid the discontinuity of the law, $\omega_i = \dot{\phi}_i$ is the rotation speed at the node i , and $\omega_{\text{Bit}} = \omega_n + \omega_{\text{motor}}$, where ω_{motor} is the downhole motor rotational speed, assumed to be constant.

Methodology

In this paper, we solve the following parameters estimation problem: Given a sequence of surface measurements $(\omega_0(t_i), \tau_m(t_i))_{1 \leq i \leq N}$ estimate the friction parameters (μ_s, μ_d) and the bit rotation speed defined by Eq.(3) and Eq.(1). To solve this problem, a deep learning approach based on Physics Informed Neural Networks (PINN) resulting in a batch method is employed. It is beneficial for solving ordinary or partial differential equations since we can integrate non-linear physical equations in the loss function to guide the training process. It allows to get physically consistent values and can handle under-determined problems.

In the following subsections, we first briefly review the basic principle of PINNs, and then discuss how to employ them to solve this problem.

Physics Informed Neural Networks

Physics Informed Neural Networks are deep learning models that are trained to solve supervised learning tasks while respecting certain laws of physics described by a set of ordinary or partial differential equations. As illustrated in Fig. 2, the PINN model proposed in this work is a fully-connected neural network that takes time as its unique input and outputs a state vector $x_\theta(t) \in \mathbb{R}^n$, where $\theta = (W, b)$ are the weights and biases of the network. The model comprises five dense layers. The hidden layers use a hyperbolic arctangent activation function, while the last layer employs a linear activation function. Each hidden layer consists of 64 neurons. The last layer, responsible for outputting $x_\theta(t)$, contains n neurons. Adam optimizer is used to minimize the loss function. The learning rate is adapted to the situation changes in the learning process using a scheduling technique which decreases it when the loss value runs into stagnation. The model is used to construct a trajectory $x_\theta(t)$ that simultaneously:

- Satisfies the nonlinear differential equation integrated in the loss function.
- Produces an output $h(\theta(t))$ that matches the measurement $y(t)$ given as input to the PINN.

These two objectives constitute the two terms of the loss function $J(\theta, p)$ that is used to train the network such that:

$$J(\theta, p) = \frac{\lambda_\eta}{N_\eta} \sum_{i=1}^{N_\eta} \|x_\theta(t_i) - f(x_\theta(t_i), u(t_i), p)\|^2 + \frac{\lambda_d}{N_d} \sum_{i=1}^{N_d} \|y(t_i) - h(x_\theta(t_i))\|^2 \quad (4)$$

The first term of the loss function is referred to as physical loss in the PINN framework, while the second term is referred to as data loss. The parameters N_d and N_η represent the number of measurements and number of collocation points for physical equations, λ_d and λ_η are the weighting coefficients of the two loss terms which may be fixed or adaptive using a self-adaptive weight technique to tune them automatically

while training (Barreau, et al., 2021; McClenny, et al., 2020). The unknown parameters p of the physical differential equation explicitly appear as arguments of the loss function, as they are treated as additional trainable variables. Automatic differentiation (Baydin, et al., 2018) is employed to compute the time derivative that appears in the physical loss.

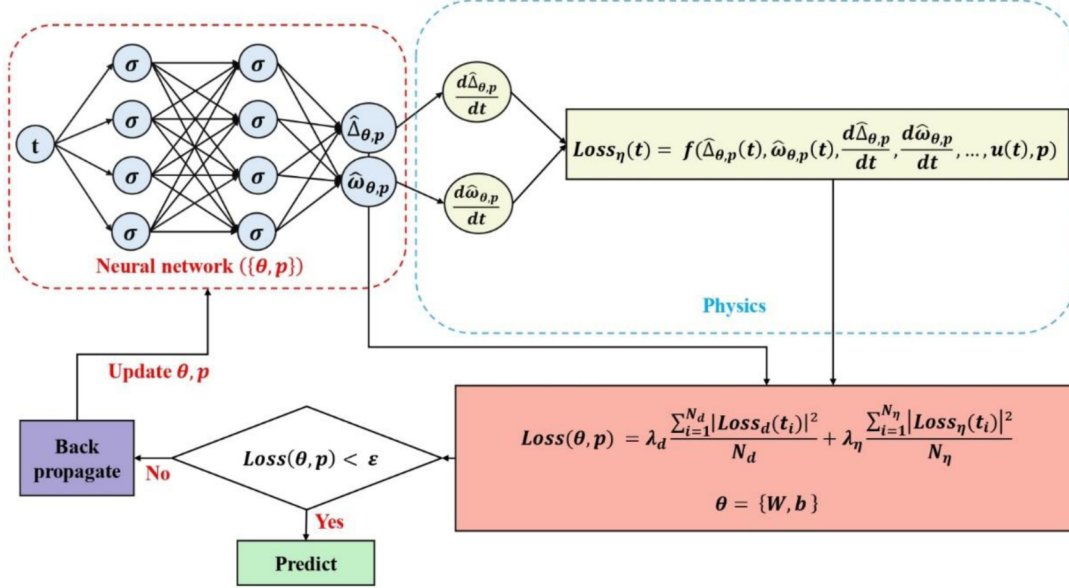


Figure 2—Schematic view of physics informed neural networks (PINNs) (Nath, et al., 2023): t is the input time; $\hat{\omega}_{\theta,p}$ and $\hat{\Delta}_{\theta,p}$ are the predicted states by the PINN, p are the predicted friction coefficients which are defined as additional trainable variables.

PINN for Friction Estimation in Drilling

To solve the defined problem, we use our framework with the torsional dynamics described by Eq.(4) with one single mass ($n = 1$). The reasons for this are two-fold:

- Measured frequencies of torsional and stick-slip oscillations are often found to be close to the estimated first natural torsional frequency of the drillstring assembly (Brett, 1992; Raissi, et al., 2017). Richard (2001) compares the stick-slip frequency to the dominant frequency of the angular velocity signal measured while rotating the bit off-bottom, which is assumed to correspond to the drillstring's natural torsional frequency. Their findings show that self-excited torsional vibrations may also occur at fundamental frequencies lower than the first natural torsional frequency, depending on operational conditions. As a result, a physical one-mass is sufficient to capture stick-slip dynamics.
- The objective of this paper is set to develop a solution capable of estimating downhole velocity and friction coefficients using low-frequency surface data. The solution is intended to be generalizable, suitable for real-time application, and directly applicable to field data. Given that surface data are typically recorded at 1 Hz, the industrial standard frequency, more complex physical models require substantially more information to produce accurate results, as additional parameters must be estimated and a larger number of equations minimized.

This yields a model with a state vector $x = (\omega_0, \Delta_0, \omega_1)$ which are respectively, the top-drive velocity, angular displacement between the top drive and the drill string, and the velocity of the entire drill string and BHA. The dynamics equation reads

$$\dot{x} = \left(\frac{\tau_m - K_{eq}\Delta_0}{I_0}, \omega_0 - \omega_p, \frac{K_{eq}\Delta_0 - C_{eq}\omega_1 - T_1^f - T_1^b}{M_{eq}} \right) \quad (5)$$

where τ_m is the surface torque, M_{eq} , C_{eq} and K_{eq} are the equivalent inertia, damping, and rigidity coefficients of the drill string as one mass. The parameter identification problem is slightly re-parameterized by defining $\{\hat{\mu}_s, \hat{r}\}$ as, respectively, the static friction coefficient and the ratio between the dynamic and the static friction coefficients ($r = \frac{\mu_d}{\mu_s}$). These parameters must satisfy the following inequalities:

$$0 \leq \hat{\mu}_d = \hat{r}\hat{\mu}_s \leq \hat{\mu}_s \leq 1 \quad (6)$$

To enforce these physical constraints within the structure of the neural network, the decision variables are chosen to be $p = \{\alpha, \beta\}$ such that $(\hat{\mu}_s, \hat{r}) = (\sigma(\alpha), \sigma(\beta))$ with σ being the sigmoid function defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \in [0, 1] \quad \forall x \in \mathbb{R}$$

This ensures that $\hat{\mu}_s = \sigma(\alpha) \in [0, 1]$ and $\hat{r} \in [0, 1] \quad \forall (\alpha, \beta) \in \mathbb{R}^2$.

Transfer Learning

With the approach of above sub-section, for any sequence of measurements $(y(t_i), u(t_i))_{i=1..N}$, the loss function (Eq. (4)) is used to find the optimal parameters (p, θ) . Thus, for each new data sequence, it is necessary to retrain the PINN entirely, which is time-consuming. To reduce the computation training cost, transfer learning strategies can be applied.

As explained in (Zhuang, et al., 2020), transfer learning focuses on transferring the knowledge from a related domain (called source domain) to improve the learning performance or minimize the number of labeled examples required in a target domain. The pre-trained source model is used as the starting point for another model with a second task of interest. The retraining can be performed on all or parts of the source model. In our case, the source model is a PINN model trained on a first set of measurements $(y(t_i), u(t_i))_{i=0..N_d}$. In the presence of a new set of measurements $(y_{new}(t), u_{new}(t))$, instead of training a new PINN to relearn the relationships between the new measurements and the new values of p_{new} and θ_{new} , we propose a transfer learning strategy, where we reuse the knowledge acquired by the source model to solve the non-linear optimization problem. The source model is fine-tuned to adapt to the new measurement sequences $(y_{new}(t), u_{new}(t))$ by retraining model weights and the additional trainable variables representing the unknown parameters p_{new} . The same optimizer and same scheduling technique for the learning rate as the source model are used. This yields a lighter computational burden while still providing a significant performance.

Numerical Results

In this section, we illustrate the performance of the proposed methodology to predict the downhole bit rotation speed and friction coefficients $\{\mu_s, \mu_d\}$, using only the available surface measurements as inputs, namely the surface velocity ω_0 and surface torque τ_m . We first evaluate the performance on simulated data from a physical model, where μ_s and μ_d are imposed. In this way we can compare the estimated parameters to the ground truth. Finally, we apply the approach to field data sampled at 1Hz from an actual drilling system.

Application on Simulated Data

A variety of synthetic data have been generated by running simulations with different values of the friction coefficients $\{\mu_s, \mu_d\} \in [0.2, 0.5] \times [0.1, 0.5]$, using the physical model mentioned in Eq. (1), where the drill

string is discretized into 15 masses₁ ($n=15$). The well is an inclined borehole with a total length of 28,645 ft, following the same trajectory as Well 2 illustrated in Fig. 6. For these simulations, a perfect mud motor imposes a minimum downhole velocity of 120 rpm. For each simulation (with given friction coefficients $\{\mu_s, \mu_d\}$), 30-second-long sequences are recorded at a frequency rate of 1 Hz for the surface velocity ω_0 , surface torque τ_m , and bit rotation speed ω_{Bit} .

For each of these sequences, the optimization problem is solved separately, i.e. a new set of parameters is found each time, without the use of the previous estimations. The training time for each simulation is around 10 min. Fig. 3 summarizes the results for the friction coefficients estimation using the generated data, with static friction shown in blue and dynamic friction in red. As shown in the figure, the predicted values are in good agreement with the true ones. Furthermore, Fig. 4 shows a comparison between the evolution of the (simulated) ground truth bit velocity ω_{Bit} with the predicted $\hat{\omega}_{Bit}$ given by the PINN, for two test simulations. We observe that the network accurately predicts the dynamics of the bit rotation speed and captures the intricate non-linear behavior of the Coulomb friction law implemented in the loss function. For these two simulations, the predicted friction coefficients are presented in Table 1. We underline that, unlike most classical numerical method for solving partial or ordinary differential equations, this prediction is obtained without any sort of time domain discretization. Rather, the time dependency of the solution is approximated by the neural network, in what could be seen as a nonlinear projection.

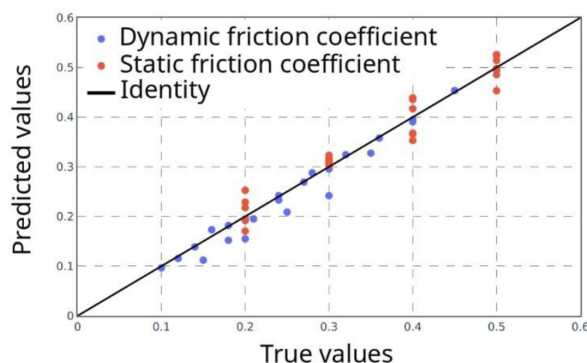


Figure 3—True and predicted values of the friction coefficients (μ_s, μ_d) when the proposed approach is applied on simulated data: Mean Absolute Percentage Error (MAPE) between the true and the predicted friction coefficients (μ_s, μ_d) and their ratio of the PINN are respectively: 0.071, 0.116 and 0.097.

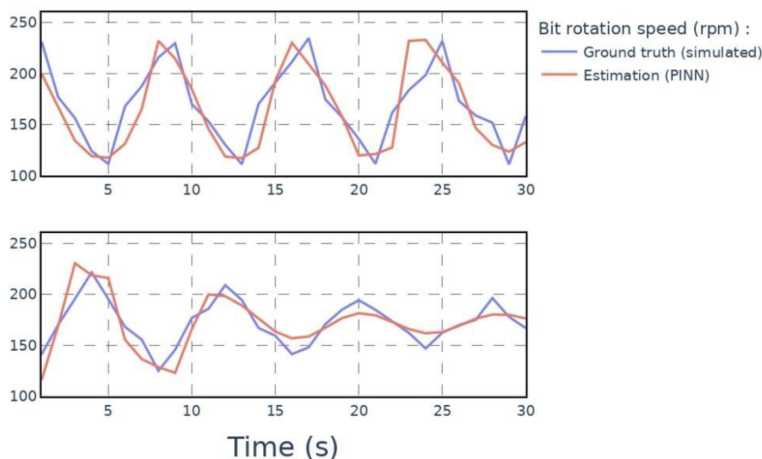


Figure 4—Simulated ω_{Bit} of two examples of the generated simulations using the n -mass physical model and predicted $\hat{\omega}_{Bit}$ given by the PINN model.

Table 1—True and predicted friction coefficients of the two examples of the generated sequences shown in Fig. 5.

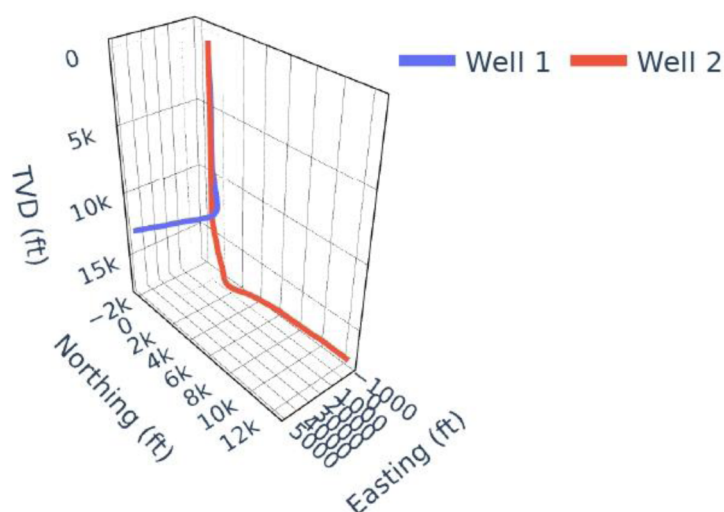
Friction coefficients	Simulation 1		Simulation 2	
	True	Predicted	True	Predicted
μ_s	0.2	0.23	0.3	0.31
μ_d	0.16	0.16	0.27	0.26

Application on Short Sequences of Field Data

In this section, we test our approach using sequences of field measured surface data $\{\omega_0, \tau_m\}$ sampled at 1 Hz. The true value of the friction coefficients is unknown. Hence, to assess the quality of the predicted friction coefficient values, we simulate the same sequence using the 15-mass physical model, described in the first section, with the predicted friction coefficients given by the PINN model, and we compare the simulated surface torque and bit rotation speed with the measured real data and the predicted ones. We test our method on two lateral test wells located in different drilling fields as outlined in Table 2 and Fig. 6.

Table 2—Well characteristics

Test well	Well 1	Well 2
Bit diameter (inch)	8 ¾	8 ¾
Drill pipe diameter (inch)	5	5 ½
Length of the well (ft)	20679	28645
Length of the lateral section (ft)	9448	10927
BHA length (ft)	110.48	152.9

**Figure 6—3D view of the two trajectories: Well 1 and Well 2.**

The approach is tested on three sequences of field data. The first is 60-second-long sequence from test well 1, and the two other sequences are 30-second-long sequences from test well 2. Table 3 and Fig. 7 depict the measured field data (used to train the PINN), and simulated surface torque, Fig. 8 shows the predicted, measured, and simulated bit rotation speed, and Table 4 presents the predicted friction coefficients, for the three test sequences. We observe that $\hat{\omega}_{\text{Bit}}$ predicted by the PINN model follows the same variations as the real measured and simulated data using the predicted friction coefficients $\hat{\mu}_s$ and $\hat{\mu}_d$. The small lag between

the simulated and the real or the predicted bit rotation speed is due to the failure to capture the real stick slip period during the simulation, specifically for sequence 1, which represents $\frac{7}{8}$ of the actual period.

Table 3—Sequences characteristics

	Sequence 1	Sequence 2	Sequence 3
Well	Test well 1	Test well 2	Test well 2
Depth (ft)	10.245	19.007	18.876
WOB (klbf)	38	50	50
Bit aggr² (lbf.ft/lbf)	0.18	0.13	0.41

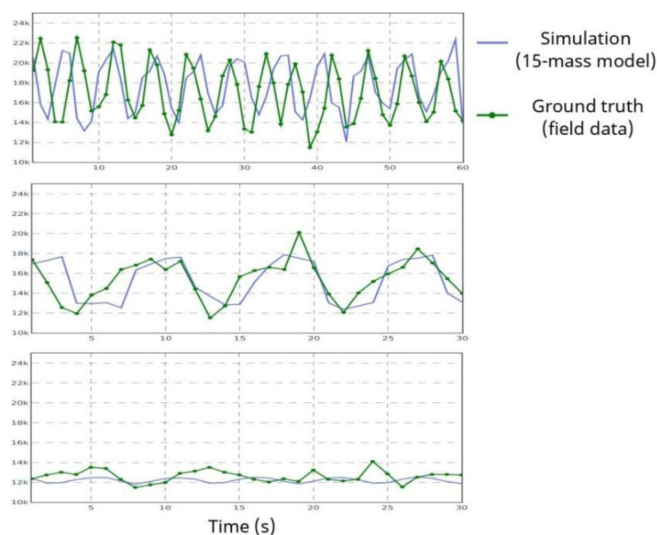


Figure 7—Comparison between the real surface torque measurements and the corresponding simulations conducted with the physical model with 15 masses using the friction coefficients predicted by the PINN for the three test sequences respectively.

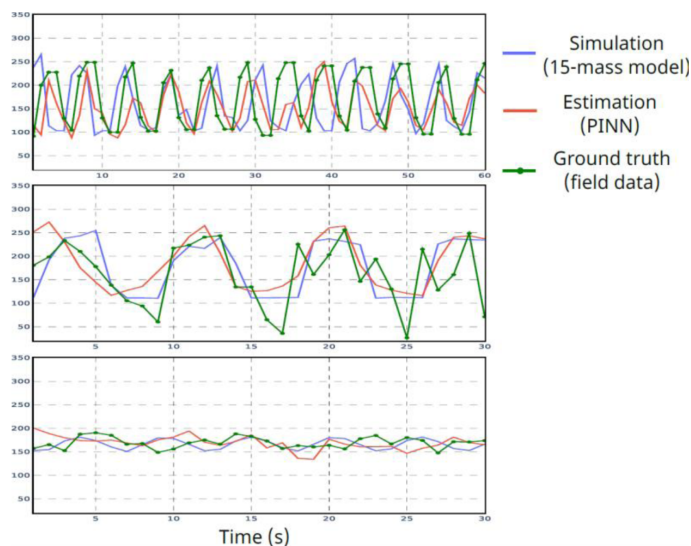


Figure 8—Comparison between the real bit velocity measurements and the corresponding simulations conducted with the physical model with 15 masses using the friction coefficients predicted by the PINN of the three test sequences respectively.

Table 4—Predicted friction coefficients by the PINN model of the three tested real data sequences. Because Sequence 2 does not feature stick-slip oscillations, the value of the static friction coefficient estimated by our method should be ignored: this parameter is not identifiable from the data of Sequence 2.

	Sequence 1	Sequence 2	Sequence 3
$\hat{\mu}_s$	0.35	0.27	(0.11)
$\hat{\mu}_d$	0.11	0.18	0.08

Application of Transfer Learning

To evaluate the effectiveness of transfer learning, we re-predict the friction coefficients and bit rotation speed of the generated test sequences explained in **Application on Simulated Data** section, using transfer learning technique instead of training a new PINN for each sequence as we did previously where the results are presented in Fig. 3. For each test sequence, transfer learning is applied. The following two test cases are considered:

- Case 1: Train a source model using a new simulated sequence with severe stick-slip.
- Case 2: Train a source model using a new simulated sequence without stick-slip.

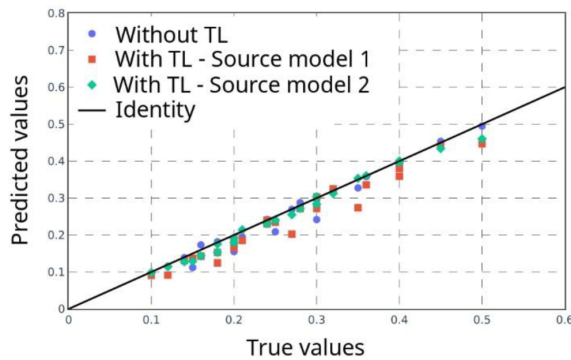
Table 5 details the true and the predicted friction coefficients of the sequences used to train the two proposed source models. For each of the two cases, and for each test sequence, transfer learning is applied. The training time is always around 96 s per sequence. The results are shown in Fig. 9, where we compare the results of the friction coefficient predictions without the application of transfer learning (same results shown in Fig. 3) and with the application of transfer learning using the two proposed source models, and Table 6 where we present mean and maximum of the mean absolute percentage error (MAPE) variations between the predicted and the true values of μ_d , μ_s , and their ratio. For the dynamic friction coefficient in Fig. 9a, the predicted values are approximately the same for the three tested PINN models and match the true values very well. For the static friction coefficient in Fig. 9b, training a new PINN model for each new sequence improves the precision of the predicted μ_s than applying transfer learning for same sequences, specifically when $\mu_s = \mu_d$. For the rest of the sequences, with or without transfer learning application, the PINN accurately predicts μ_s . By adjusting only some parameters of the source model, we decrease the training time of the PINN model without losing in the accuracy of the predicted friction coefficients for most of the test sequences.

Table 5—True and predicted friction coefficients of the two trained source models.

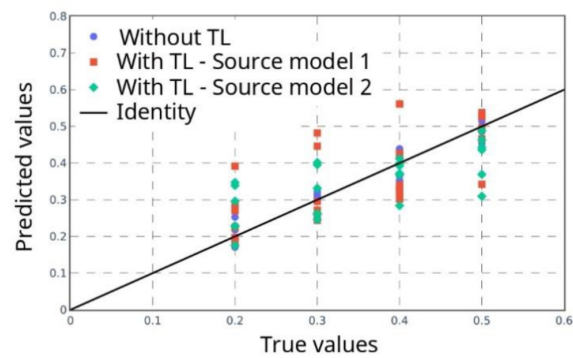
	True μ_s	Predicted μ_s	True μ_d	Predicted μ_d
Case 1	0.2	0.2	0.16	0.15
Case 2	0.1	0.13	0.09	0.08

Table 6—Mean and maximum Mean Absolute Percentage Error (MAPE) between the true and the predicted friction coefficients and their ratio of the PINN without transfer learning application, and with the application of transfer learning using the two proposed source models.

	μ_s		μ_d		$\frac{\mu_d}{\mu_s}$	
	Mean	Max	Mean	Max	Mean	Max
Without TL	0.07	0.11	0.06	0.25	0.09	0.17
With TL - Source model 1	0.22	0.33	0.11	0.30	0.19	0.38
With TL - Source model 2	0.20	0.38	0.05	0.14	0.19	0.26



(a) Dynamic friction coefficient



(b) Static friction coefficient

Figure 9—True and predicted friction coefficients of the generated test sequences using the PINN without the application of transfer learning (same results as Fig. 3, and with the application of transfer learning tested on two different source models. Source model 1: source model trained on a sequence with severe stick-slip; Source model 2: source model trained on a sequence without stick-slip.

Application on Long Sequences of Field Data

To further illustrate its capabilities, the PINN is tested on the entire well drilling data of test well 1 described in Table 3. This includes six days of drilling data. The surface data is divided into 30 s sequences, and for each sequence, friction coefficients ($\hat{\mu}_s, \hat{\mu}_d$) and bit rotation speed $\hat{\omega}_{Bit}$ are predicted and compared to the real measurements. Fig. 10 presents the comparison results for the test sequence exhibiting severe stick-slip in the test well. The surface torque, surface velocity, surface estimated TOB and downhole measurements of TOB variations are presented along with the real bit rotation speed measurements, which is compared with the predicted $\hat{\omega}_{Bit}$ by the PINN. The real bit rotation speed measurements are represented by three curves: maximum, minimum, and mean values derived from high-frequency downhole sensor data sampled at 80 Hz over a 2.56 s interval. As shown in Fig. 10 the PINN accurately reproduces the bit rotation speed measurements, including the maximum, minimum, and mean values using only surface measurements.

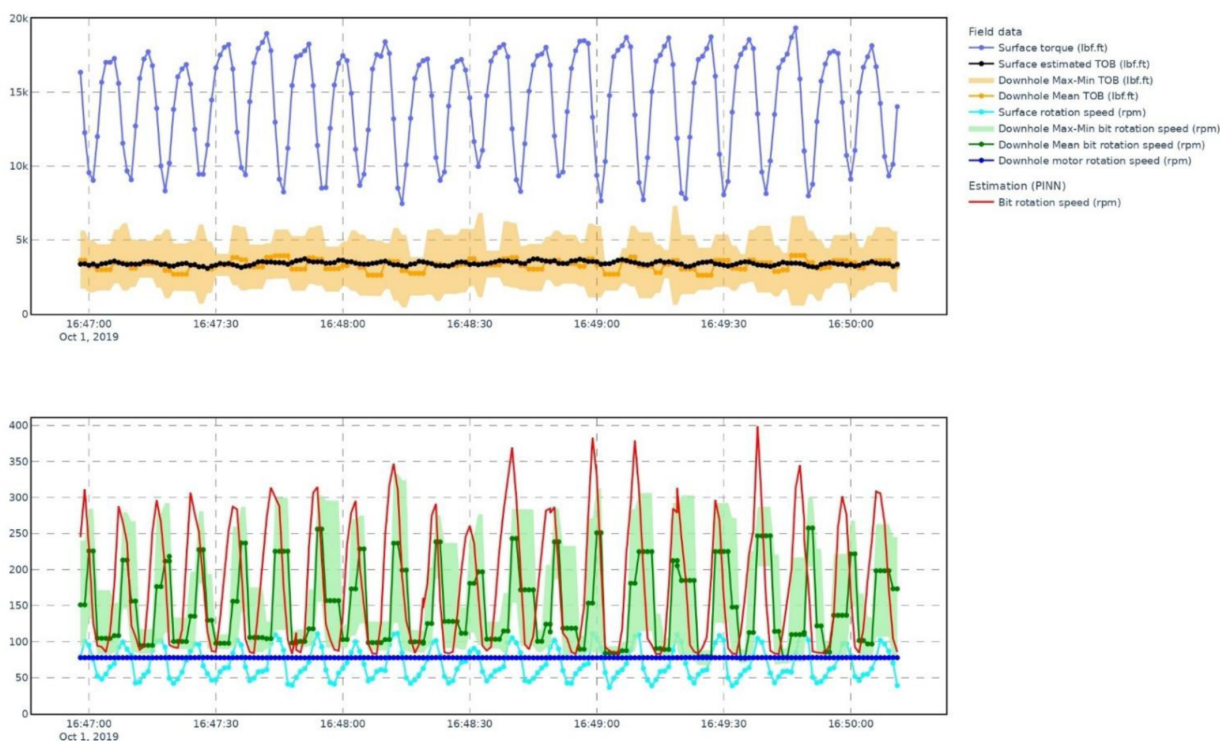


Figure 10—Comparison between the actual bit rotation speed measurements and the predicted $\hat{\omega}_{Bit}$ values by the PINN, using the real surface torque and surface velocity as inputs for test Well 1. The flow rate remains constant at 266gpm throughout the sequence. The real bit rotation speed is measured using downhole sensor and obtained after the well is drilled.

To demonstrate the performance of the PINN on extended sequences, Fig. 11 presents the calculated moving maximum, minimum and average of both, downhole measurements for bit rotation speed, and the PINN predicted bit rotation speed, along with the moving average of downhole motor rotation speed, and Table 7 represents the Mean Absolute Error (MAE) and the Pearson correlation coefficient between the field and predicted bit rotation speed. As we can see, the predicted $\hat{\omega}_{Bit}$ reproduces successfully the variations of the downhole bit rotation speed only 1 Hz surface data as input.

Table 7—Mean Absolute Error (MAE) and Pearson correlation coefficient (r) between the moving max, mean and min of true and predicted bit rotation speed of the two test wells.

	MAE	r
Moving max	6.458	0.994
Moving mean	1.262	0.998
Moving min	4.402	0.982

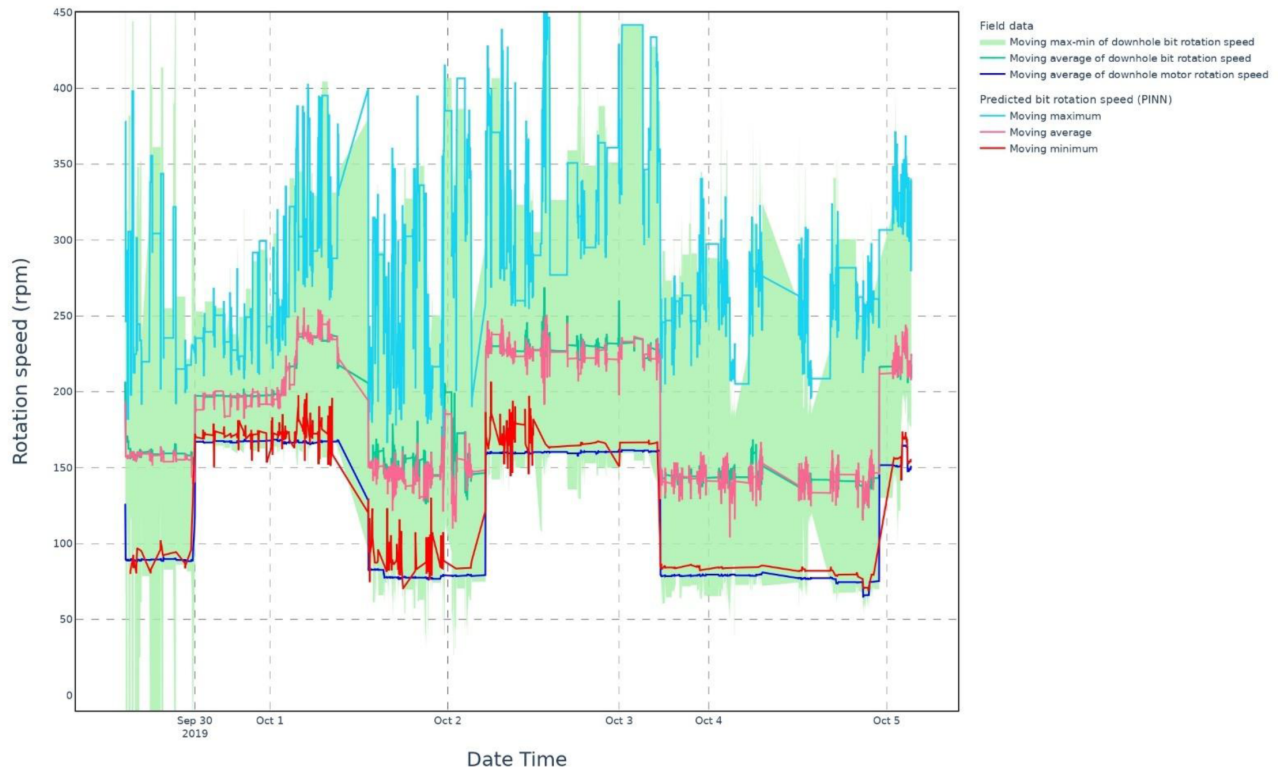


Figure 11—The variations of downhole bit rotation speed measurements and the predicted $\hat{\omega}_{Bit}$ values by the PINN, using the real surface torque and surface velocity as inputs for test well 1 over six drilling days, are shown. The moving minimum, maximum and average of both, the field bit rotation speed measurements and the PINN predicted bit rotation speed, along with the moving average of the downhole motor rotation speed, are calculated within a 120 s window.

Fig. 12 displays the evolution of friction coefficients predicted by the PINN using surface measurements as input for test well 1. For the static friction coefficient $\hat{\mu}_s$, only values observed when the sequence features rotation speeds below the characteristic rotary speed $\omega_c = 30$ rpm, as defined in Eq.(3), are shown. Indeed, the static friction coefficient has more effect for low rotation speeds. Given the uncertainty of the real friction coefficients, analyzing their variations is quite challenging. Consequently, we chose to compare the dynamic friction coefficient predicted by the PINN with the one estimated using the torque and drag (T&D) model (see e.g (Aarsnes, et al., 2019; Menand, et al., 2006)). T&D computations are run in drilling every 1 ft for different dynamic friction coefficients using WOB and TOB from surface measurements. WOB is estimated based on the difference between off-bottom and on-bottom hookload measurements, and TOB is calculated using differential pressure, derived from the difference between off-bottom and on-bottom standpipe pressure measurements. These computations yield different curves of surface torque (one for each friction coefficient) function of bit depth (see Fig. 13). To obtain the dynamic friction coefficient corresponding to the time-based data, first, surface torque measurements are averaged every 60 s to reduce the torque dynamic variations and keep only a steady-state answer that is more prone to be compared with the static T&D results. Then, each point of the obtained surface torque measurements is placed in the graph of Fig. 13 and a friction coefficient is computed by interpolation (or extrapolation).

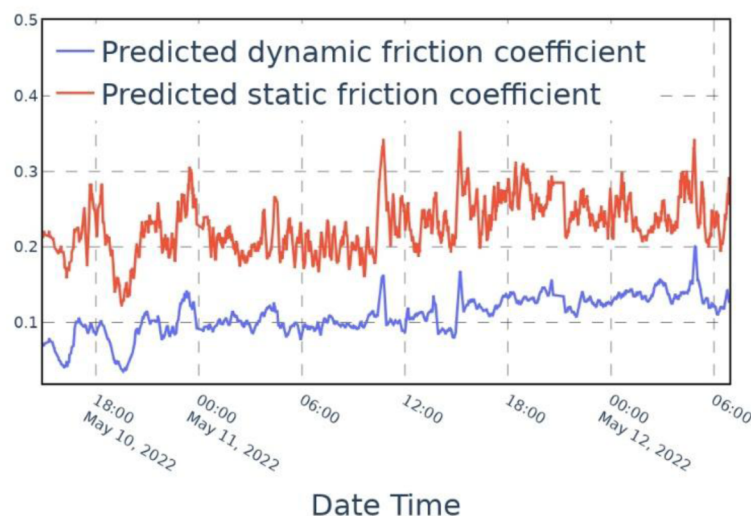


Figure 12—The PINN predicted friction coefficients of the test well 1. Only the static friction coefficient values detected when the bit rotation speed is less than three times the characteristic rotary speed ω_c are displayed.

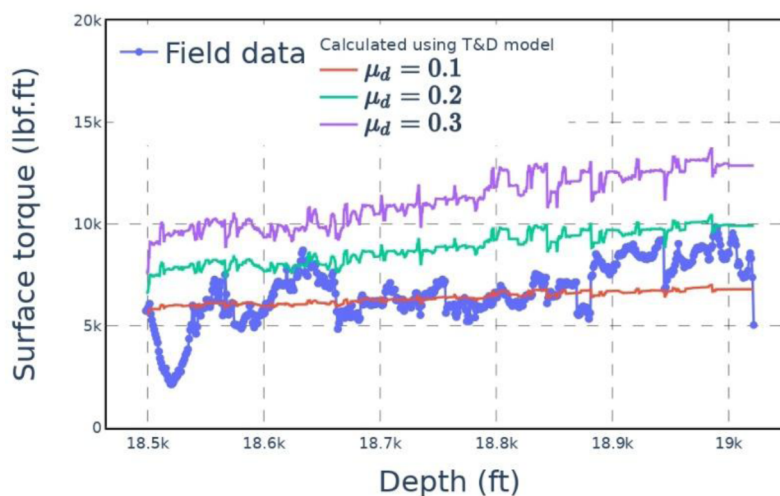


Figure 13—Surface torque variations over depth for the test well 1: Comparison between actual measurements and T&D model calculations using various dynamic friction coefficient values.

Fig. 14 shows the dynamic friction coefficient as predicted by both the PINN and T&D models. The two curves exhibit similar patterns but with a varying offset between them. This variation is attributed to the different friction laws used in each model, where T&D model operates under the assumption that μ_d is equal to μ_s , whereas the PINN predicts both μ_d and μ_s using the friction law mentioned in Eq.(3). As conclusion, it is encouraging to observe that the variation of the dynamic friction coefficient predicted by the PINN aligns with that of the T&D model, which is the traditional method used in the industry for estimating dynamic friction coefficients. However, a key advantage of the PINN is its ability to predict μ_s in addition to μ_d , which is crucial for evaluating transitional periods of the drill string, such as during startup and shutdown.

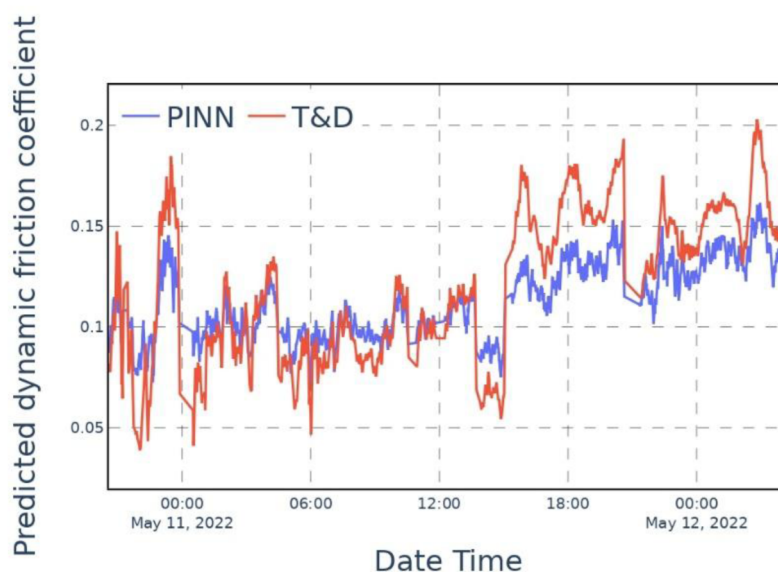


Figure 14—Predicted dynamic friction coefficients by the PINN and T&D model variations over time of the test well 1.

Conclusion

This work presents a PINN-based approach to estimate the bit rotation speed and both static and dynamic friction coefficients between the drill string and the wellbore in a torsional drilling system, using only 1 Hz surface measurements (torque and velocity). Unlike traditional physical models that require prior knowledge of friction parameters, the PINN is trained exclusively on surface data, making it generalizable and applicable to any well. The method is validated on synthetic datasets of varying sequence lengths (30 s and 60 s), achieving accurate predictions of both the bit speed and friction coefficients. It is further tested on real field data, where it successfully predicts the bit rotation speed, with results aligning well with downhole measurements.

Because ground-truth friction coefficients are unavailable in the field data, evaluating the predicted coefficients remains challenging. Additionally, the PINN must be retrained when the input or field data changes. To address this, a transfer learning strategy is implemented, significantly reducing training time by a factor of six, while maintaining high accuracy in predicting friction coefficients. Comparative evaluations demonstrate that dynamic friction predictions from the PINN aligned well with those from the T&D model, the latter could not estimate the static friction coefficient, highlighting a key advantage of the PINN approach.

Directions for future work include integrating a hydraulic physical model into the PINN for the downhole motor and testing it using high-frequency surface data to replicate bit-sticking events. Additionally, implementing a PINN with a physical model that incorporates an arbitrary number of masses in the loss function. However, this requires the use of higher-frequency data that would contain information on higher vibration modes. From our experience, calibrating the loss function weights for a dynamical system with more than four states is tedious. In the present case, it would seem better suited to implementing a PINN relying on the partial differential equations (PDE) (Dao, et al., 2019). Moreover, it is important to analyze the factors that contribute to the variations in the predicted friction coefficients of the PINN using surface measurements and try to distinguish between them. Finally, the proposed approach is generic and could be applied to other parameter estimation problems, in the drilling context and in other settings. To evaluate its merits, a benchmark study comparing it to other generic grey-box system identification solvers could be performed.